Assign8_7323

March 5, 2026

[153]:
```
#Dushyant krishna sharma
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt


df = pd.read_csv('titanic.csv')
df.head()
```

[153]:
```
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name     Sex   Age  SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th…  female  38.0      1
2                             Heikkinen, Miss. Laina  female  26.0      0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                           Allen, Mr. William Henry    male  35.0      0

   Parch            Ticket     Fare Cabin Embarked
0      0         A/5 21171   7.2500   NaN        S
1      0          PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0            113803  53.1000  C123        S
4      0            373450   8.0500   NaN        S
```

[154]:
```
df.describe()
```

[154]:
```
       PassengerId    Survived      Pclass         Age       SibSp  \
count   891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std     257.353842    0.486592    0.836071   14.526497    1.102743
min       1.000000    0.000000    1.000000    0.420000    0.000000
```

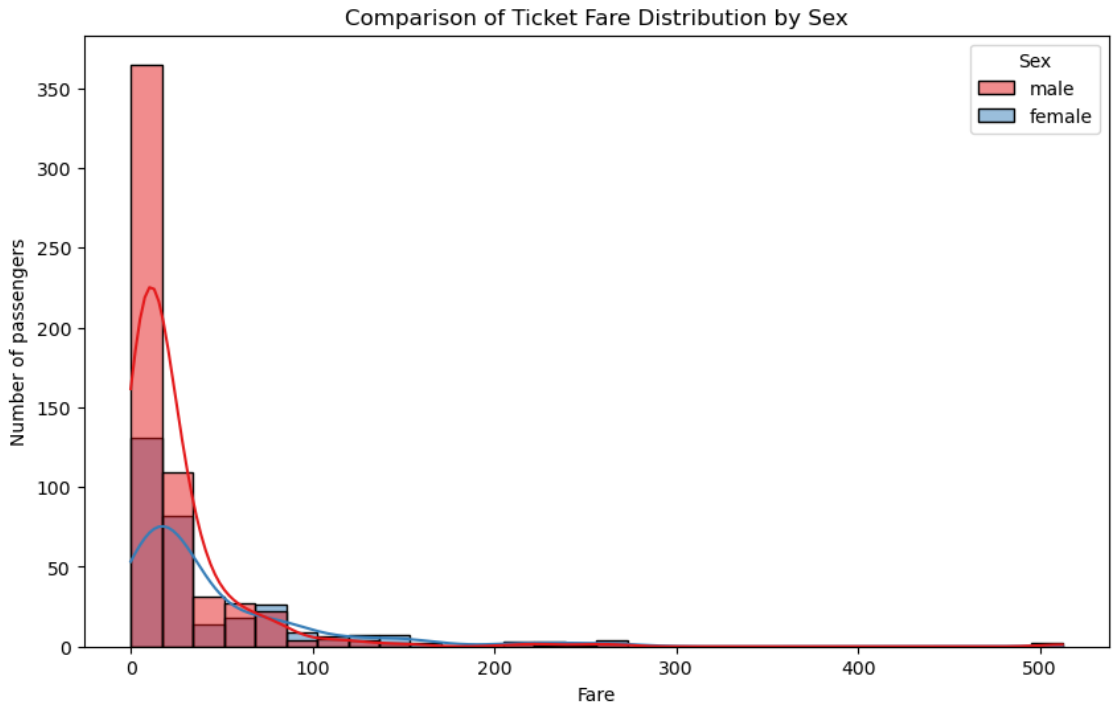
```
25%     223.500000    0.000000    2.000000   20.125000    0.000000
50%     446.000000    0.000000    3.000000   28.000000    0.000000
75%     668.500000    1.000000    3.000000   38.000000    1.000000
max     891.000000    1.000000    3.000000   80.000000    8.000000

            Parch        Fare
count  891.000000  891.000000
mean     0.381594   32.204208
std      0.806057   49.693429
min      0.000000    0.000000
25%      0.000000    7.910400
50%      0.000000   14.454200
75%      0.000000   31.000000
max      6.000000  512.329200
```

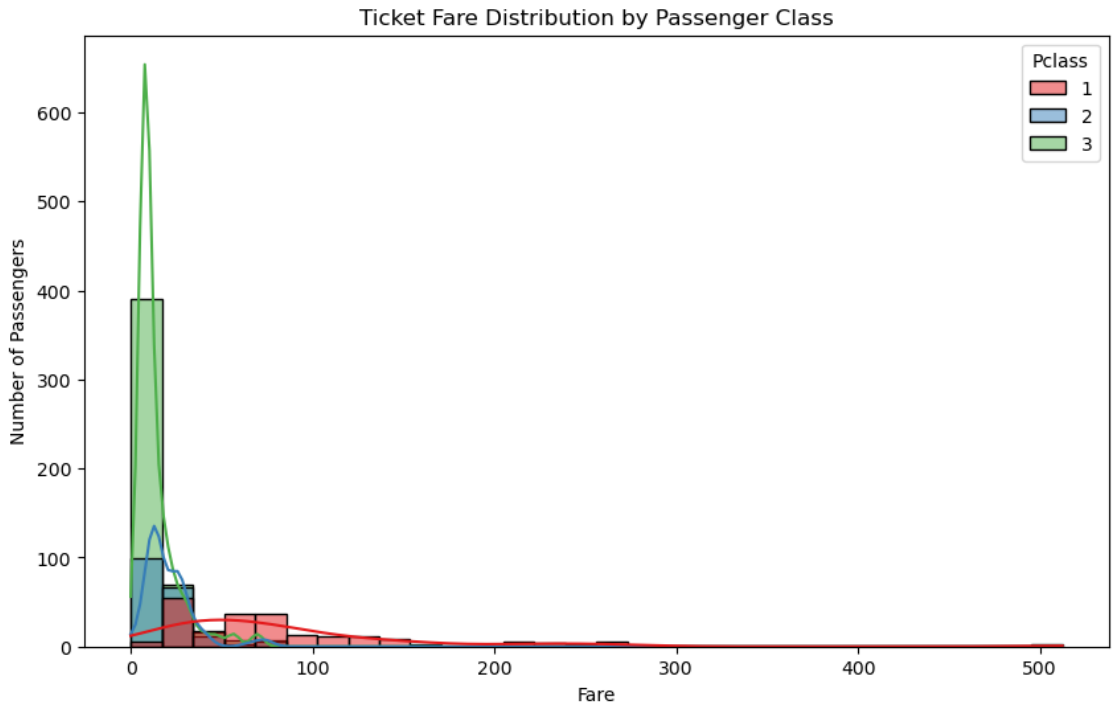
[155]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

[156]:
```
plt.figure(figsize=(10,6))
sns.histplot(data=df, x='Fare', hue='Sex', bins=30, kde=True, palette='Set1')
plt.title('Comparison of Ticket Fare Distribution by Sex')
plt.xlabel('Fare')
plt.ylabel('Number of passengers')
plt.show()
```

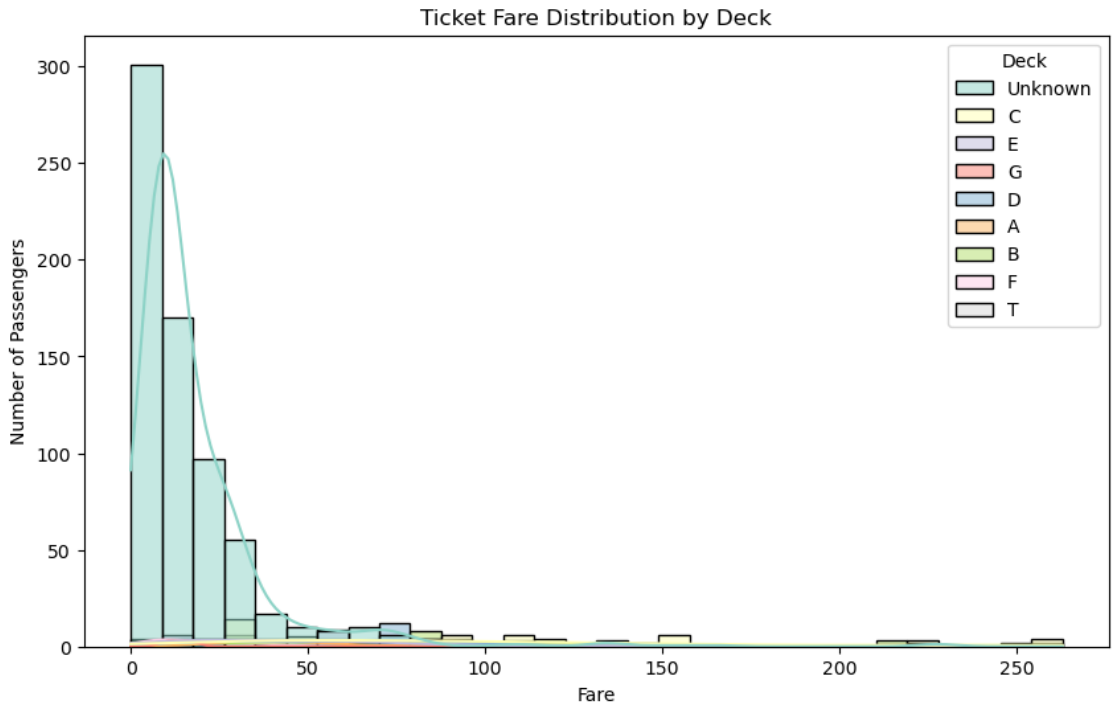


[157]:
```
plt.figure(figsize=(10,6))
sns.histplot(data=df, x='Fare', hue='Pclass', bins=30, kde=True, palette='Set1')
plt.title('Ticket Fare Distribution by Passenger Class')
plt.xlabel('Fare')
plt.ylabel('Number of Passengers')
plt.show()
```



[158]:
```
df['Cabin'] = df['Cabin'].fillna('Unknown')
df['Deck'] = df['Cabin'].apply(lambda x: x[0] if x != 'Unknown' else 'Unknown')
df_hist = df[df['Fare'] < 500]

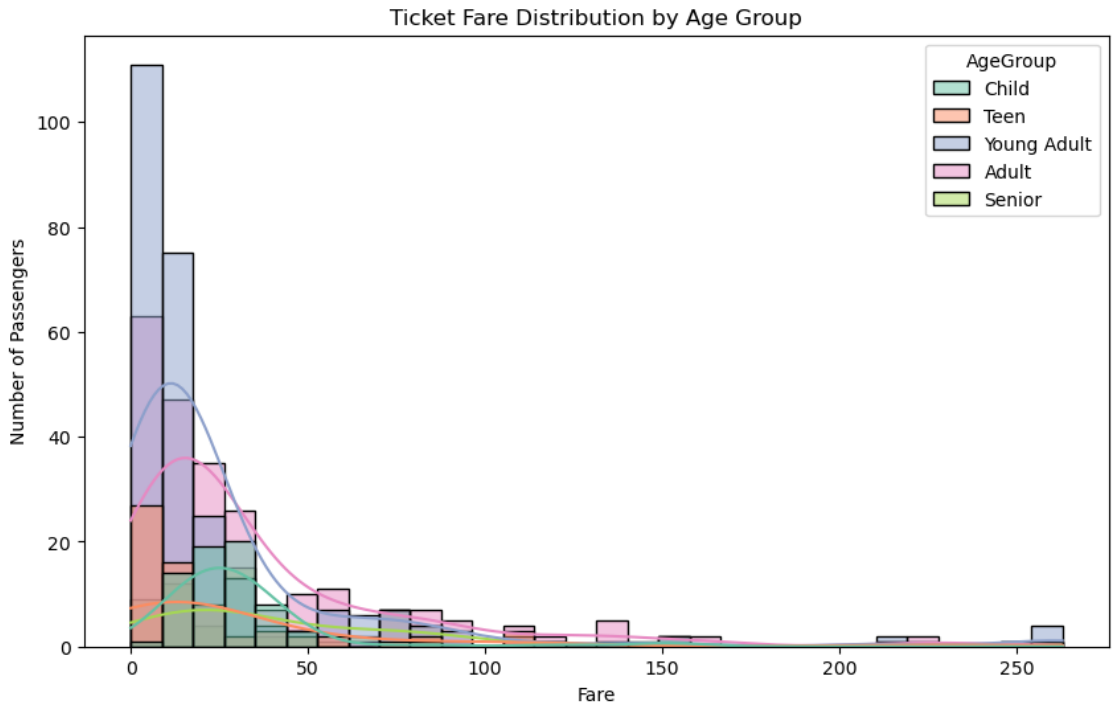
plt.figure(figsize=(10,6))
sns.histplot(data=df_hist, x='Fare', hue='Deck', bins=30, kde=True,␣
 ↪palette='Set3')
plt.title('Ticket Fare Distribution by Deck')
plt.xlabel('Fare')
plt.ylabel('Number of Passengers')
plt.show()
```



[159]:
```
df_clean = df.dropna(subset=['Age', 'Fare'])
df_clean = df_clean[df_clean['Fare'] < 500]

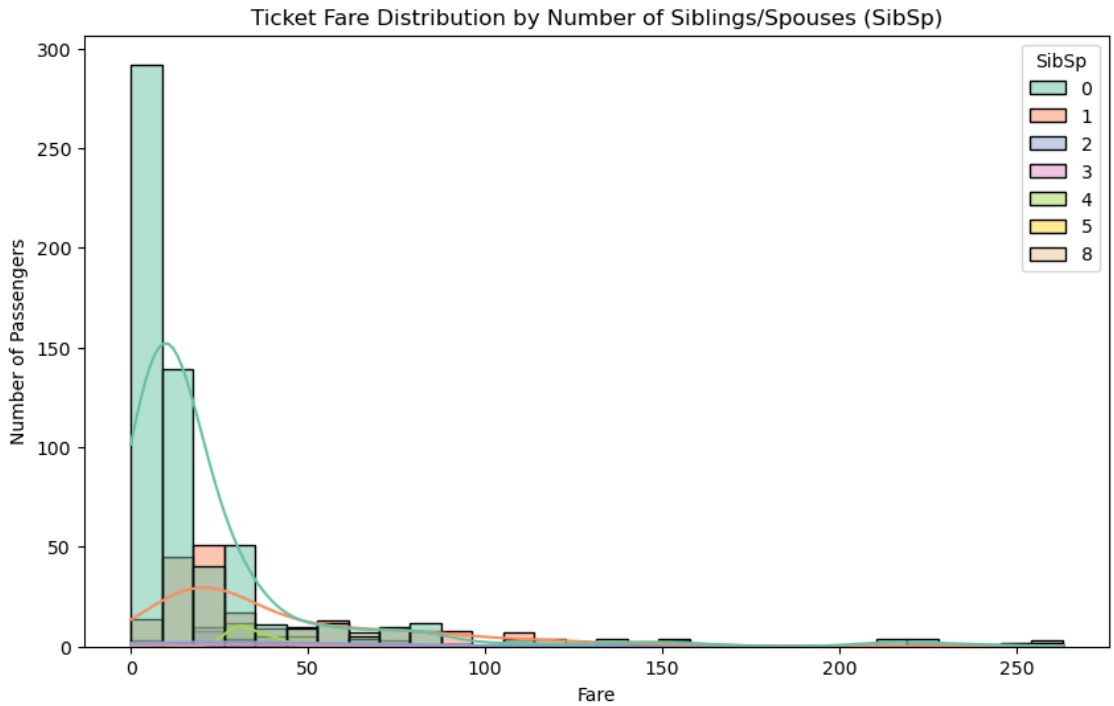

df_clean['AgeGroup'] = pd.cut(df_clean['Age'], bins=[0,12,18,30,50,80],␣
↪labels=['Child','Teen','Young Adult','Adult','Senior'])


plt.figure(figsize=(10,6))
sns.histplot(data=df_clean, x='Fare', hue='AgeGroup', bins=30, kde=True,␣
↪palette='Set2')
plt.title('Ticket Fare Distribution by Age Group')
plt.xlabel('Fare')
plt.ylabel('Number of Passengers')
plt.show()
```

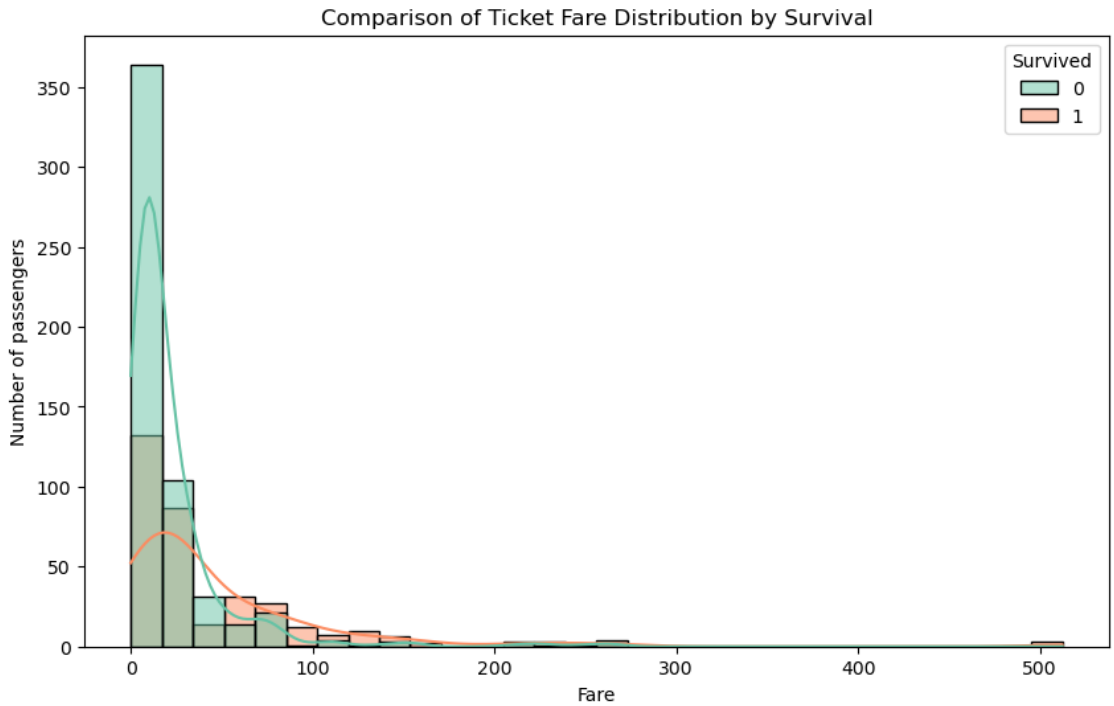


[160]:
```
df_clean = df[df['Fare'] < 500]

plt.figure(figsize=(10,6))
sns.histplot(data=df_clean, x='Fare', hue='SibSp', bins=30, kde=True,␣
↪palette='Set2')
plt.title('Ticket Fare Distribution by Number of Siblings/Spouses (SibSp)')
plt.xlabel('Fare')
plt.ylabel('Number of Passengers')
plt.show()
```



[161]:
```
plt.figure(figsize=(10,6))
sns.histplot(data=df, x='Fare', hue='Survived', bins=30, kde=True,␣
 ↪palette='Set2')
plt.title('Comparison of Ticket Fare Distribution by Survival')
plt.xlabel('Fare')
plt.ylabel('Number of passengers')
plt.show()
```


Comparison of Ticket Fare Distribution by Survival
Survived
0
1
Number of passengers
Fare
0
50
100
150
200
250
300
350
0
100
200
300
400
500

```
[ ]:
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